👑 CODERKING STUDIO

Official Technical Roadmaps & Architecture Guides

AI & LLM ARCHITECTURE

DeepSeek R1: Pure Reinforcement Learning Blueprint

How a $6M model rivaled OpenAI o1 via Group Relative Policy Optimization (GRPO) without human supervision.

🧠 1. The GRPO Paradigm

Traditional LLMs rely heavily on expensive Supervised Fine-Tuning (SFT) with millions of labeled human answers.

DeepSeek R1 Innovation: Uses Group Relative Policy Optimization (GRPO) directly on the base model with zero human labeled steps.

- **Rule-Based Rewards:** LeetCode/compiler pass + exact math solution accuracy.
- **Zero Critic Model:** Evaluates groups of answers against the mean, cutting GPU VRAM by 50%.

💡 2. The 'Aha Moment' (Self-Correction)

During extended RL training, the model naturally developed an internal reflection chain-of-thought:

```
Thinking Process:
1. Try formula A -> contradiction det
ected
2. Wait, let me reconsider approach B
3. Re-verifying steps 1 to 4...
4. Correct final output confirmed.
```

The model spontaneously taught itself to double-check its work before outputting answers!

⚡ 3. Multi-Head Latent Attention (MLA)

Standard Multi-Head Attention requires massive KV cache memory during inference.

- **MLA Compression:** Compresses KV states into a low-dimensional latent space.
- **Result:** 93% smaller KV cache per user request, allowing 10x larger batch sizes on a single GPU node.

💻 4. Run 100% Free Offline (Ollama)

Run the distilled DeepSeek R1 reasoning models directly on your laptop with zero monthly subscription:

```
# 1. Install Ollama
curl -fsSL https://ollama.com/instal
l.sh | sh

# 2. Run 7B or 14B model locally
ollama run deepseek-r1:7b

# 3. Test reasoning prompt
>>> Solve: How many 'r's are in straw
berry?
```

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Official GitHub: **github.com/shubh74/coderking-studio**